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BANGALORE · INDIA

AGRIMITRA : AI-Powered Smart Farming Assistant

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MISSION

CHRIST is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

VISION

Excellence and Service

CORE VALUES

Faith in God | Moral Uprightness
Love of Fellow Beings
Social Responsibility | Pursuit of Excellence

Agenda

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Abstract

AgriMitra is an AI-powered smart farming assistant designed to provide personalized agricultural support to farmers. The platform integrates IoT-based environmental monitoring, weather data, intelligent irrigation recommendations, machine learning-based crop recommendation and rotation planning, crop price prediction, and an AI-powered agricultural assistant. Real-time farm context such as soil moisture, temperature, humidity, crop growth stage, weather conditions, and irrigation status is used to generate personalized recommendations. The platform also supports multilingual AI interaction through voice input and output, making agricultural guidance more accessible to farmers.

Introduction

- Agriculture remains the primary livelihood for a large share of India's rural population, yet most smallholder farmers still rely on manual observation rather than real-time data for irrigation, crop selection, and market decisions.
- Existing smart-irrigation systems, reviewed in the next section, automate watering but stop short of combining sensor data with AI-driven crop guidance, price forecasting, and natural-language interaction.
- AgriMitra is proposed as an AI-powered, multilingual farming assistant that unifies IoT-based environmental monitoring, machine-learning crop and irrigation recommendations, image-based price prediction, and a voice-enabled conversational assistant into a single platform.

Literature Review

1) Darshana et al. (2025) – Smart Irrigation System

- Uses soil moisture sensors, temperature sensor, relay, and Arduino/Node-based automatic pump control.
- Advantage: Reduces water wastage and enables irrigation.
- Limitation: No IoT connectivity, no remote monitoring, no AI support.

2) El Mezouari et al. (2022) – Smart Irrigation System

- Uses Arduino UNO, soil moisture sensor, temperature sensor, water-level sensor, and relay for automatic irrigation.
- Advantage: Efficient water usage and automatic monitoring.
- Limitation: Relies on serial communication and lacks cloud connectivity/intelligent decision-making.

3) Badreideen et al. (2022) – IoT-Based Smart Irrigation System

- Uses soil moisture, temperature, humidity, rain sensors, along with IoT connectivity/mobile application.
- Advantage: Real-time monitoring, remote control, and weather-related information.
- Limitation: No AI assistant or multilingual voice interaction.

Gap Identified

- **Sensor-and-relay-based systems (Darshna et al., 2025)** automate irrigation but provide no IoT connectivity, remote monitoring, or AI support.
- **Arduino-based systems with local sensing (El Mezouari et al., 2022)** rely on serial communication only, lacking cloud connectivity and intelligent decision-making.
- **IoT-enabled systems (Badreideen et al., 2022)** add real-time monitoring and remote control but still offer no AI assistant or multilingual voice interaction for farmers.
- **No reviewed system combines IoT sensing, ML-based crop and irrigation intelligence, image-based price prediction, and multilingual voice assistance in one accessible platform** — the AgriMitra system is designed to close this gap.

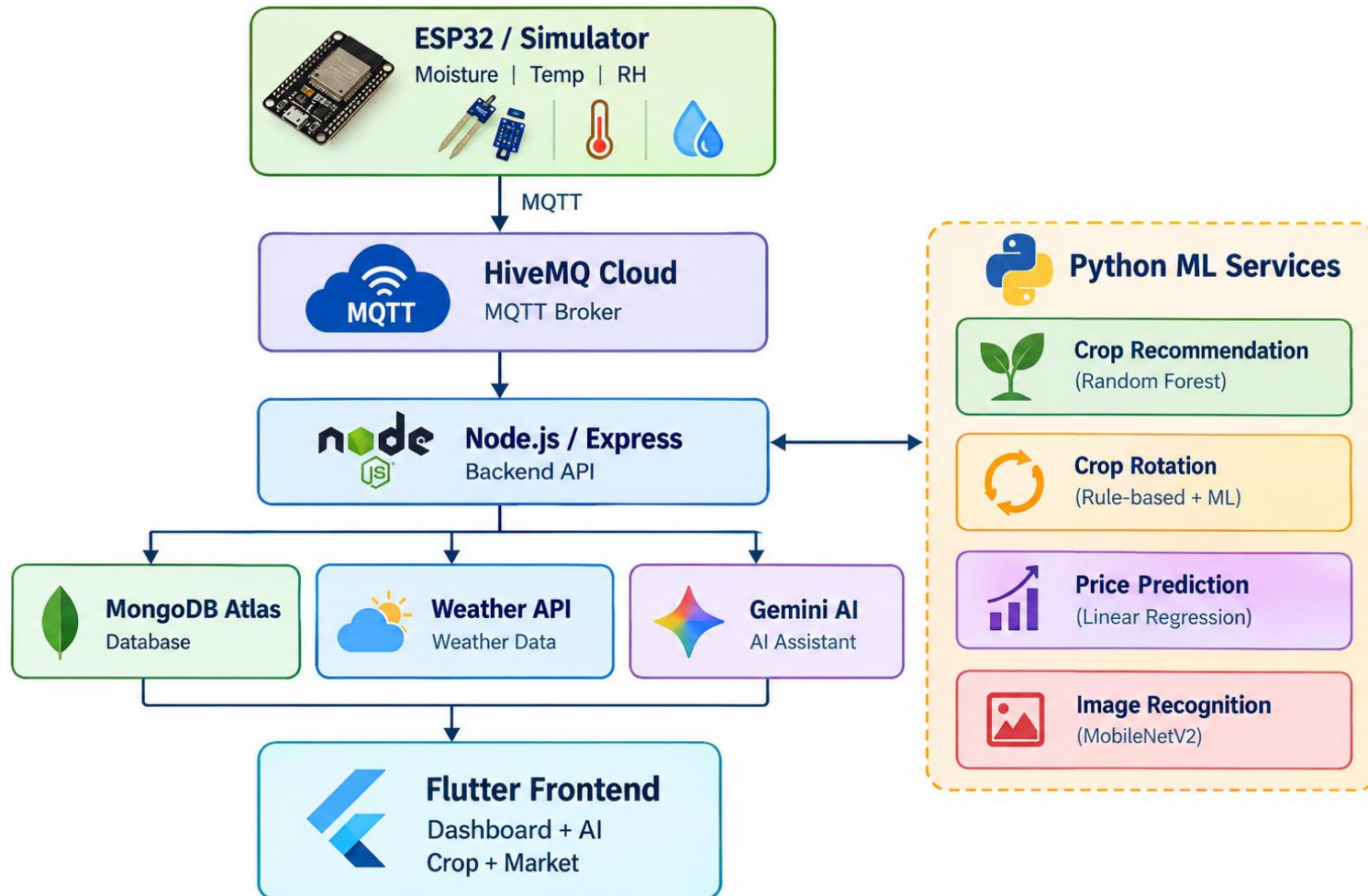
Problem Statement

- Farmers often lack intelligent tools to plan suitable crop rotations and farming activities based on soil conditions, previous crops, and seasonal factors.
- Farmers face difficulty in identifying crops from images and obtaining reliable estimates of their current and future market prices.
- Farmers need an accessible system for real-time farm monitoring and personalized agricultural guidance through simple voice-based interaction.

Proposed Model

- **Crop Planning & Recommendation:** Uses soil conditions, previous crop history, and seasonal factors to recommend suitable crops and farming activities through ML and rule-based decision-making.
- **Crop Identification & Price Prediction:** Uses **MobileNetV2** to identify crops from images and ML-based market analysis to estimate current and future prices.
- **IoT-Based Farm Monitoring:** Collects real-time environmental data such as soil and weather conditions using IoT sensors.
- **Multilingual AI Assistant:** A **Gemini-powered voice assistant** provides farmers with real-time farm information, recommendations, and personalized guidance in multiple languages.
- **Integrated Architecture:** Combines **IoT sensing, ML models, crop identification, price prediction, weather information, and conversational AI** into a single platform for intelligent agricultural decision-making.

Design - System Architecture



Design - Modules

Crop Rotation & Scheduling

- Recommends suitable crop sequences based on **soil conditions and previous crops**.
- Suggests appropriate **farming activities and schedules**.
- Uses **machine learning and rule-based analysis** for crop planning.

Image-Based Crop Price Prediction

- Identifies the crop from an **uploaded image using MobileNetV2**.
- Retrieves relevant **market price data**.
- Predicts **current and future crop price trends** using historical data.

Implementation -Hardware and Software Requirements

Component	Purpose
ESP32	IoT sensor connectivity
Capacitive Soil Moisture Sensor	Soil moisture monitoring
DHT22	Temperature & humidity
Relay Module	Planned pump switching
12V DC Pump	Proposed irrigation actuator
Water Level Sensor	Planned dry-run protection
Power Supply	Development/testing

Implementation - Software Stack

Layer	Technology
Frontend	Flutter / Dart
Backend	Node.js / Express
Database	MongoDB Atlas
IoT Communication	MQTT / HiveMQ Cloud
AI	Google Gemini 2.5 Flash
Crop ML	Python / Flask / Random Forest
Price ML	Python / Flask
Image Recognition	MobileNetV2
Weather	OpenWeatherMap
Market Data	AGMARKNET / data.gov.in
Authentication	JWT + bcrypt
Voice	speech_to_text + flutter_tts

Implementation - AI & ML Components

AI Component

- **Gemini 2.5 Flash**
- Uses real-time farm context, weather information, crop details, and Irrigation information to provide personalized agricultural assistance.
- **Voice Assistance**
- Speech-to-text and text-to-speech enable multilingual voice interaction.

ML Components

Application	Model / Algorithm
Crop Recommendation	Random Forest Classifier
Crop Rotation	Hybrid ML + Rule-based Approach
Crop Image Recognition	MobileNetV2
Price Prediction	Linear Regression / Trend Forecasting
Irrigation	Hargreaves $ET_0 + K_c$ + Rule-based Decision Logic

Implementation - Screenshots

< Crop Rotation

Get AI-powered crop rotation advice based on your soil and climate.

Soil Type

▲ Red Soil

Previous Crop

🔄 Groundnut

🕒 27.9°C, 63% humidity - ~500mm rainfall (Estimated)

🔄 Get Recommendation

Recommendations

#1 Finger Millet Overall Fit: 65%

Consumes nitrogen enriched by Groundnut

Model Confidence

27%

Soil Suitability

100%

Rotation Benefit

67%

• Different plant family from Groundnut (Legume) — breaks pest and disease buildup

• Nitrogen behavior: Depletor — consumes nitrogen enriched by Groundnut

• Soil suitability: Well-suited (S1, score 100/100) - Red soil well-drained semi-arid ideal for finger millet; traditional crop in red soil regions

• Needs 51-100mm water — region averages ~500mm (South Indian semi-arid average), good drainage recommended

#2 Sesame Overall Fit: 53%

Diversifies pest exposure while matching soil moisture needs

#3 Rice Overall Fit: 40%

Consumes nitrogen enriched by Groundnut

Price Prediction

Estimate current mandi prices for your produce, with optional photo detection.

Crop
tomato

State
Andhra Pradesh

District (optional)
Any district

Upload a photo of your produce (optional)
Choose Image

₹ Get Price Estimate

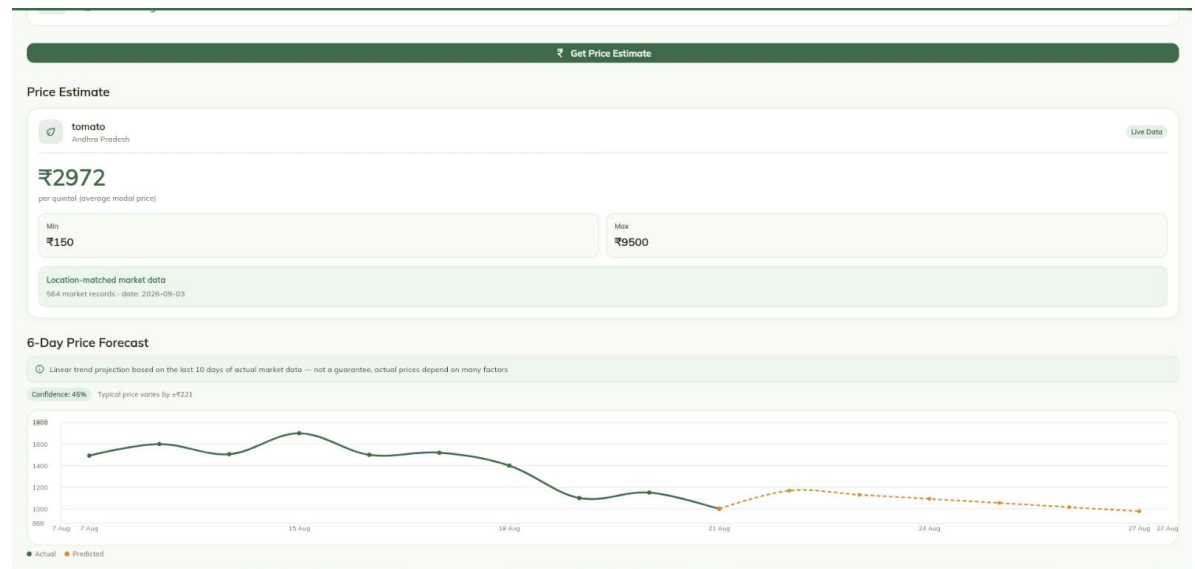
Price Estimate

tomato
Andhra Pradesh
Live Data

₹2972
per quintal (average modal price)

Min
₹150
Max
₹9500

Location-matched market data
564 market records - date: 2026-09-03



Result and Discussion

- The backend (Node.js/Express, MongoDB Atlas, JWT/bcrypt authentication) and the MQTT-based IoT pipeline are functional; sensor data is currently simulated pending real ESP32 hardware integration.
- The Random Forest crop-recommendation model achieves close to 99.55% test accuracy on Kaggle dataset, and Hargreaves ETo-based irrigation engine produces weather-grounded scheduling recommendations.
- The Flutter web and mobile frontend is wired to live backend data, with a working dashboard, an auto-generated farm calendar, and a throttled smart-alert system.
- The Gemini 2.5 Flash-powered AI assistant maintains multi-turn conversation history and responds using real-time farm context; voice input/output work in English , Tamil , Telugu ,Malayalam and Hindi.

Conclusion

- AgriMitra demonstrates that IoT-based monitoring, ML-driven crop and irrigation recommendations, image-based crop prediction, and multilingual voice interaction can be combined into a single platform for Indian farmers.
- Testing so far shows high-accuracy crop-recommendation model, a working weather-grounded irrigation engine, and a functional multilingual conversational assistant, directly addressing the gaps found in the reviewed prior systems.
- The project validates that combining rule-based and machine-learning approaches — chosen for viable-feasibility — is a practical strategy where data constraints limit purely ML-based solutions.

Future Scope

- Complete real ESP32 hardware integration to replace the currently simulated sensor data.
- Extend voice support to Kannada through the Bhashini API, and verify Tamil text-to-speech output across devices.
- Strengthen ML/DS credibility with a confusion matrix, multi-model comparison (XGBoost, SVM) for crop recommendation, and add Isolation Forest anomaly detection for sensor data.
- Optionally extend into crop disease detection, evaluated on PlantVillage and PlantDoc datasets to address the known lab-to-field generalization gap.

Future Scope – Project Roadmap

Phase	Duration	Deliverable
Requirement analysis & literature review	Week 1–2	Finalized scope, reviewed prior work
Hardware setup & sensor integration	Week 3–4	ESP32 + sensors + relay + pump
Backend & database architecture	Week 5–6	MQTT pipeline, MongoDB, REST APIs
Irrigation logic (ET _o engine)	Week 7–8	Weather-grounded irrigation logic
Crop recommendation ML model	Week 9	Trained and saved model (99.55% accuracy)
Conversational AI + voice assistant	Week 10–12	Gemini-based, multilingual voice IO
Authentication, alerts, farm calendar	Week 13–14	Secure authentication system, sensor alerts, scheduling
Hardware-software full integration	Week 15–16	Real ESP32 replacing simulated sensor data
Testing, refinement, documentation	Week 17–18	End-to-end testing, report writing
Final review & viva preparation	Week 19+	Presentation, demo readiness

Proposed Solution

- The platform provides machine learning-based crop recommendations and crop rotation plans by considering soil suitability, previous crops, and relevant agricultural factors.
- The system identifies crops from images and provides crop price estimates and future price trends using market data from AGMARKNET.
- AgriMitra monitors real-time soil moisture, temperature, humidity, and weather conditions to provide farmers with up-to-date information about their farm environment.
- The system determines irrigation requirements using Hargreaves ET_0 , crop coefficient (K_c), crop growth stage, soil moisture, and rainfall probability.
- A Gemini 2.5 Flash-powered AI assistant uses real-time farm data to provide personalized agricultural guidance through multilingual text and voice interaction.

References

- El Mezouari, A. E., Fazziki, A. E., and Sadgal, M., “Smart Irrigation System,” *IFAC-PapersOnLine*, vol. 55, no. 10, pp. 3298–3303, 2022. DOI: 10.1016/j.ifacol.2022.10.125.
- Badreleen et al., “Smart irrigation system based on IoT and machine learning,” *Smart Agricultural Technology*, vol. 3, 2022. DOI: 10.1016/j.s.2022.100147.
- AI4Bharat — IndicTrans: Open-source speech models for Indian languages, IIT Madras.
- Kaggle Crop Recommendation Dataset — N-P-K, temperature, humidity, pH, rainfall crop label dataset.

THANK YOU